

July 15, 2001. Five days later, astronauts Carl Walz and Daniel Bursch were the first to perform a spacewalk through the new airlock.

Pirs

The 16-foot-long, 3,500-kg Pirs Docking Compartment was attached to the bottom, Earth-facing port of the Zvezda on September 16, 2001. Pirs has two primary functions: it serves as a docking port for the docking of transport and cargo vehicles to the ISS and as an airlock for the performance of spacewalks by two crewmembers using Russian Orlan spacesuits. In addition, it can transport fuel to and from the fuel tanks of a docked Progress or Soyuz spacecraft to Zvezda or Zarya.

Harmony

For the next six years, the ISS didn't see any modular enhancements. Several trusses were added in this period to improve functions, but the next major addition would happen only in October 2007, when the US-made Harmony module was brought on board. The installation of NASA's Harmony increased the living and working space inside the station to approximately 500 cubic metres. It also allowed the addition of international laboratories from Europe and Japan to the station, providing a passageway between three station science experiment facilities: the US Destiny Laboratory, the upcoming Kibo Japanese Experiment Module, and the soon-to-be-launched European Columbus Laboratory. The aluminium node is 7.2 metres long and 4.4 metres in diameter, weighing approximately 14,288 kilograms.

Columbus

The ESA's largest single contribution to the space station, the research laboratory module Columbus, was attached in February 2008. Columbus is about 23 feet long and 15 feet wide, allowing it to hold 10 "racks" of experiments, each approximately the size of a phone booth. Each rack provides independent controls for power and cooling, as well as communication links to earthbound controllers and researchers. These links allow scientists all over Europe to participate in their own experiments in space from several user centres and, in some cases, even from their own work locations. The Columbus laboratory's flexibility provides room for the researchers on the ground, aided by the station's crew, to conduct thousands of experiments in life sciences, materials sciences, fluid physics and other research in a weightless environment not possible on Earth. In addition, the station crew can conduct experiments outside the module within the vacuum